

Notes on "Numerical Methods for Optimal Transportation"

Applied Math Seminar

1 Kantorovich Formulation of Optimal Transport Problem

Suppose (a, b) are histograms in the simplices $\Sigma_n \times \Sigma_m$, $C \in \mathbb{R}^{m \times n}$ is the cost matrix, define the set of coupling measures

$$U(a, b) := \{P \in \mathbb{R}_+^{m \times n} | P\mathbf{1}_m = a \text{ and } P^T\mathbf{1}_n = b\}, \quad (1)$$

it is bounded and defined by $n + m$ equality constraints, and therefore is a convex polytope (the convex hull of a finite set of matrices).

Kantorovich's optimal transport problem now reads

$$L_C(a, b) := \min_{P \in U(a, b)} \langle C, P \rangle = \sum_{i,j} P_{i,j} C_{i,j}. \quad (2)$$

This is a linear program, and as is usually the case with such programs, its optimal solutions are not necessarily unique.

The Kantorovich problem (2) is a constrained convex minimization problem, and as such, it can be naturally paired with a so-called dual problem, which is a constrained concave maximization problem. The following fundamental proposition explains the relationship between the primal and dual problems.

Proposition 1. *The Kantorovich problem (2) admits the dual*

$$L_C(a, b) = \max_{(f, g) \in R(C)} \langle f, a \rangle + \langle g, b \rangle, \quad (3)$$

where the set of admissible dual variables is

$$R(C) := \{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m | \forall (i, j) \in [n] \times [m], f_i \oplus g_j \leq C_{i,j}\}, \quad (4)$$

where $f \oplus g = f\mathbf{1}_m^T + \mathbf{1}_n g^T$.

Primal (2) and dual (3) problems can be solved independently to obtain optimal primal P^* and dual (f^*, g^*) solutions. They have some relationship as below:

Proposition 2. *Let P^* and (f^*, g^*) be optimal solutions for the primal (2) and dual (3) problems. Then, for any $(i, j) \in [n] \times [m]$, $P_{i,j}^*(C_{i,j} - (f_i^* + g_j^*)) = 0$ holds. In other words, if $P_{i,j}^* > 0$, then necessarily $f_i^* + g_j^* = C_{i,j}$; if $f_i^* + g_j^* < C_{i,j}$ then necessarily $P_{i,j}^* = 0$.*

Proposition 3. *If P and (f, g) are complementary and feasible solutions for the primal (2) and dual (3) problems, respectively, then P and (f, g) are both primal and dual optimal.*

As problem (2) can be treated as a traditional linear programming problem, it can be solved by a set of optimization algorithms used to solve LP problems, like simplex and interior-point method. And a method called **network simplex** is especially designed for solving problem like ot, which can be viewed as a variant of dual simplex method. Orlin [1997] was the first to prove the polynomial time complexity of the network simplex. Tarjan [1997] provided shortly after an improved bound in $O((n+m)nm \log(n+m) \log((n+m)\|C\|_\infty))$ which relies on more efficient data structures to help select pivoting edges.

2 Sinkhorn's Algorithm

To alleviate the computational burden brought by traditional methods, a large number of efficient computational tools have been developed in the recent decade. One major class of approaches is called the regularization-based method, which solves an entropy-regularized OT problem instead of the original one (Cuturi, 2013).

This regularization has several important advantages, which make it a very useful tool:

- The minimization of the regularized problem can be solved using a simple alternate minimization scheme;
- The scheme translates into iterations that are simple matrix-vector products, making them particularly suited to execution of GPU;
- For some applications, these matrix-vector products do not require storing an $n \times m$ cost matrix, but instead only require access to a kernel evaluation;
- The resulting approximate distance is smooth with respect to input histogram weights and positions of the Diracs and can be differentiated using automatic differentiation.

2.1 Entropic Regularization

The discrete entropy of a coupling matrix is defined as

$$H(P) = - \sum_{i,j} P_{i,j} (\log(P_{i,j}) - 1), \quad (5)$$

with an analogous definition for vectors, with the convention that $H(a) = -\infty$ if one of the entries a_j is 0 or negative. The function H is 1-strongly concave, because its Hessian is $\partial^2 H(P) = -\text{diag}(1/P_{i,j})$ and $P_{i,j} \leq 1$. The idea of the entropic regularization of optimal transport is to use $-H$ as a regularizing function to obtain approximate solutions to the original transport problem (2):

$$L_C^\epsilon(a, b) := \min_{P \in U(a,b)} \langle C, P \rangle - \epsilon H(P). \quad (6)$$

Since the objective is an ϵ -strongly convex function, Problem (6) has a unique optimal solution.

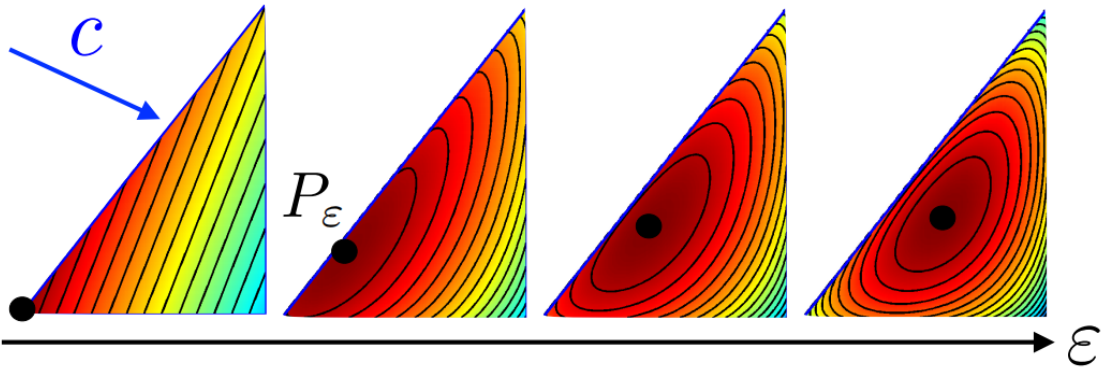


Figure 1: Impact of ϵ on the optimization of a linear function on the simplex.

Proposition 4 (Convergence with ϵ). *The unique solution P_ϵ of (6) converges to the optimal solution with maximal entropy within the set of all optimal solutions of the Kantorovich problem, namely*

$$P_\epsilon \xrightarrow{\epsilon \rightarrow 0} \arg \min_P \{-H(P) : P \in U(a, b), \langle P, C \rangle = L_C(a, b)\}, \quad (7)$$

so that in particular

$$L_C^\epsilon(a, b) \xrightarrow{\epsilon \rightarrow 0} L_C(a, b).$$

One also has

$$P_\epsilon(a, b) \xrightarrow{\epsilon \rightarrow \infty} a \otimes b = ab^T. \quad (8)$$

Formula (7) states that for a small regularization ϵ , the solution converges to the maximum entropy optimal transport coupling. A key insight is that, as ϵ increases, the optimal coupling becomes less and less sparse, which in turn has the effect of accelerating computational algorithms.

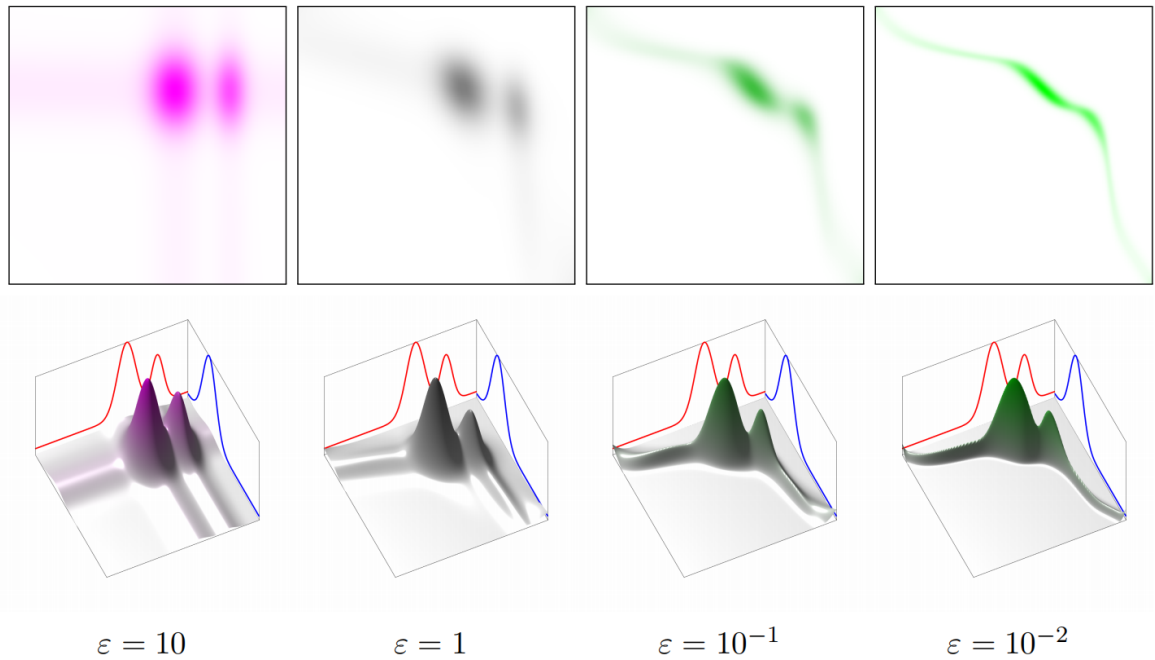


Figure 2: Impact of ϵ on the couplings between two 1-D densities, illustrating Proposition 4.

Define the KL divergence between couplings as

$$KL(P|K) := \sum_{i,j} P_{i,j} \log \left(\frac{P_{i,j}}{K_{i,j}} \right) - P_{i,j} + K_{i,j}, \quad (9)$$

the unique solution P_ϵ of (6) is a projection onto $U(a, b)$ of the Gibbs kernel associated to the cost matrix C as

$$K_{i,j} := e^{-\frac{C_{i,j}}{\epsilon}}.$$

So we have

$$P_\epsilon = \text{Proj}_{U(a,b)}^{\text{KL}}(K) := \arg \min_{P \in U(a,b)} KL(P|K) \quad (10)$$

2.2 Sinkhorn's Algorithm and Its Convergence

It can be shown that the solution of (6) has a specific form, which can be parameterized using $n + m$ variables.

Proposition 5. *The solution of (6) is unique and has the form*

$$\forall (i, j) \in [n] \times [m], \quad P_{i,j} = u_i K_{i,j} v_j \quad (11)$$

for two (unknown) scaling variable $(u, v) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$.

Proof. Introducing two dual variables $f \in \mathbb{R}^n, g \in \mathbb{R}^m$ for each marginal constraint, the Lagrangian of (6) reads

$$L(P, f, g) = \langle P, C \rangle - \epsilon H(P) - \langle f, P \mathbf{1}_m - a \rangle - \langle g, P^T \mathbf{1}_n - b \rangle.$$

First order conditions then yield

$$\frac{\partial L(P, f, g)}{\partial P_{i,j}} = C_{i,j} + \epsilon \log(P_{i,j}) - f_i - g_j = 0,$$

so an optimal P coupling to the regularized problem, is in the expression $P_{i,j} = e^{f_i/\epsilon} e^{-C_{i,j}/\epsilon} e^{g_j/\epsilon}$, which can be rewritten in the form provided above using nonnegative vectors u and v . $\square$

The factorization of the optimal solution exhibited in Equation (11) can be conveniently rewritten in matrix form as $P = \text{diag}(u) K \text{diag}(v)$. The variables (u, v) must therefore satisfy the following nonlinear equations which correspond to the mass conservation constraints inherent to $U(a, b)$:

$$\text{diag}(u) K \text{diag}(v) \mathbf{1}_m = a, \quad \text{and} \quad \text{diag}(v) K^T \text{diag}(u) \mathbf{1}_n = b. \quad (12)$$

These two equations can be further simplified as

$$u \odot (Kv) = a \quad \text{and} \quad v \odot (K^T u) = b, \quad (13)$$

where $\odot$ corresponds to entrywise multiplication of vectors. An intuitive way to handle these equations is to solve them iteratively, by modifying first u so that it satisfies the left-hand side and then v to satisfy its right-hand side. These two updates define Sinkhorn's algorithm,

$$u^{(\ell+1)} := \frac{a}{Kv^{(\ell)}} \quad \text{and} \quad v^{(\ell+1)} := \frac{b}{K^T u^{(\ell+1)}}, \quad (14)$$

initialized with an arbitrary positive vector $v^{(0)} = \mathbf{1}_m$.

Algorithm 1 SINKHORN(K, a, b, δ)

- 1: Initialize: $v^{(0)} = \mathbf{1}_m$,
 - 2: **for** $\ell = 0, 1, \dots$ **do**
 - 3: $u^{(\ell+1)} = \frac{a}{Kv^{(\ell)}} \quad \text{and} \quad v^{(\ell+1)} = \frac{b}{K^T u^{(\ell+1)}}$;
 - 4: **if** $\|u^{(\ell+1)} - u^{(\ell)}\|_1 + \|v^{(\ell+1)} - v^{(\ell)}\|_1 \leq \delta$ **then**
 - 5: $exit$;
 - 6: **end if**
 - 7: **end for**
 - 8: Output $P_\epsilon = \text{diag}(u^{(\ell+1)}) K \text{diag}(v^{(\ell+1)})$.
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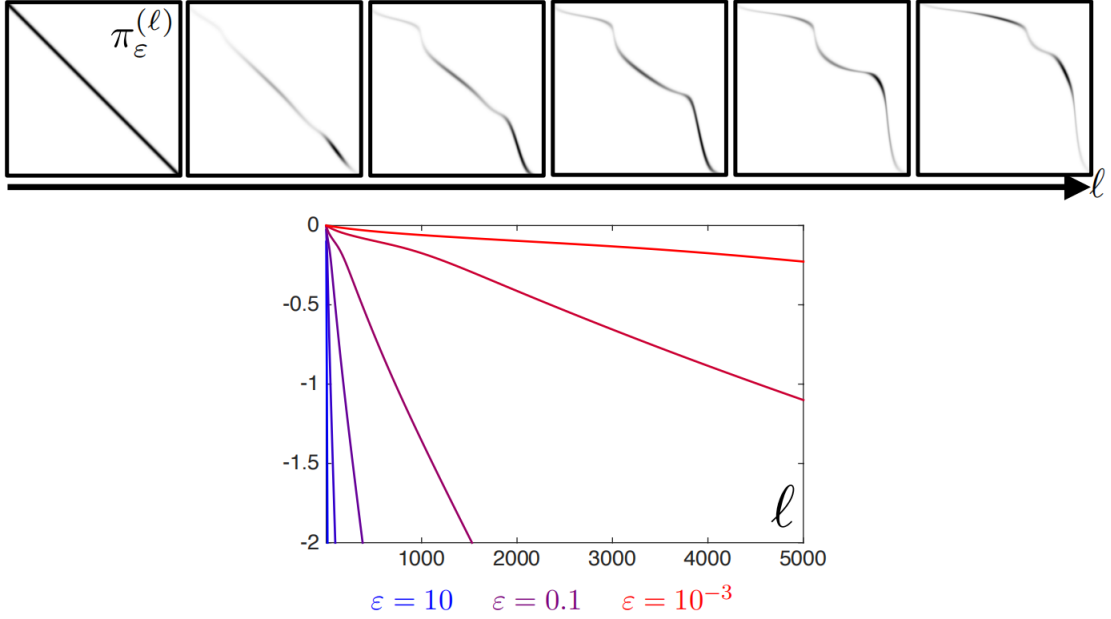


Figure 3: Evolution of $\pi_\epsilon^{(\ell)} = \text{diag}(u^{(\ell)})K\text{diag}(v^{(\ell)})$ and impact of ϵ the convergence rate of Sinkhorn, as measured in log-term of marginal constraint violation

Overall Complexity. Assuming $n = m$ for the sake of simplicity, the computational cost of traditional methods is of the order $O(n^3 \log(n))$, while the regularized OT problem can be solved using the Sinkhorn algorithm in $O(Ln^2)$ time, where L is the number of iterations. Altschuler et al. [2017] showed that by setting $\epsilon = \frac{\tau}{4 \log(n)}$, $O(\|C\|_\infty^3 \log(n) \tau^{-3})$ Sinkhorn iterations (with an additional rounding step to compute a valid coupling $\hat{P} \in U(a, b)$) are enough to ensure that $\langle \hat{P}, C \rangle \leq L_C(a, b) + \tau$. This implies that Sinkhorn computes a τ -approximate solution of the unregularized OT problem in $O(n^2 \log(n) \tau^{-3})$ operations.

The Sinkhorn algorithm suffers from numerical overflow when the regularization parameter ϵ is small compared to the entries of the cost matrix C .

3 Importance Sparsification for Sinkhorn Algorithm

Despite the wide application, the time and memory requirements of the Sinkhorn algorithm **grow quadratically with** n , which hinders its broad applicability to many large-scale optimal transport problems. To address the computational bottleneck, many efficient variants of the Sinkhorn algorithm have been proposed. For example, in contrast to the scheme of Sinkhorn that updates all rows and columns of the transport plan at each step, the variants including greedy Sinkhorn (**Greenkhorn**) (Altschuler et al., 2017; Lin et al., 2022), randomized Sinkhorn (**Randkhorn**) (Lin et al., 2019a), and screening Sinkhorn (**Screenkhorn**) (Alaya et al., 2019) only **update partial row(s) or column(s)** in each iteration, based on different selection criteria. These variants have been shown to converge faster in practice, making them appealing for large-scale applications.

A randomized sparsification variant of the Sinkhorn algorithm, called **Spar-Sink** for OT problems is introduced today. This method constructs a sparsified kernel matrix $\tilde{K}$ by carefully sampling $s = o(n^2)$ elements from K and setting the remaining ones to zero. $\tilde{K}$ is then leveraged and sparse matrix multiplications are used to accelerate the iterations in the Sinkhorn algorithm, reducing the computational cost from $O(n^2)$ to $O(s)$ per iteration.

3.1 Main Algorithm

The matrix sparsification is implemented via the Poisson sampling framework. Compared to the other commonly used subsampling technique, sampling with replacement, Poisson sampling has a **higher approximation accuracy** in some situations and is **more convenient to implement** in distributed systems.

The importance sampling technique can be described as follows: to approximate the summation $\mu = \sum_{i=1}^N f_i$ with $f_i \geq 0$, we assign each $i \in [N]$ a probability $q_i \geq 0$ such that $\sum_{i=1}^N q_i = 1$, and then sample a subset of size $s (< N)$, $\{i_t\}_{t=1}^s$, from $[N]$ based on the probabilities $\{q_i\}_{i=1}^N$. The summation then can be approximated by $\mu \approx \sum_{t=1}^s f_{i_t} / (sq_{i_t})$. When f_i 's are known, the optimal sampling probability q_i in terms of variance-reduction is proportional to f_i . When unknown, a popular surrogate is using a proper upper bound of f_i , denoted by $q'_i (i \in [N])$, as the (un-normalized) sampling probability.

According to the principle of Poisson sampling, the $\tilde{K}$ is formulated as follows: given a subsampling parameter $s < n^2$ and a set of sampling probabilities $\{p_{ij}\}_{(i,j) \in [n][n]}$ such that $\sum_{i,j} p_{i,j} = 1$, we construct $\tilde{K}$ by selecting and rescaling a small fraction of elements from K and zeroing out the remaining elements, i.e.

$$\tilde{K}_{i,j} = \begin{cases} K_{i,j}/p_{i,j}^*, & \text{with prob. } p_{i,j}^* = \min(1, sp_{i,j}) \\ 0, & \text{otherwise.} \end{cases} \quad (15)$$

The rescaling factor $p_{i,j}^*$ ensures that the sparsified kernel matrix $\tilde{K}$ is unbiased. $\mathbb{E}(\text{nnz}(\tilde{K})) = \sum_{i,j} p_{i,j}^* \leq s \sum_{i,j} p_{i,j} = s$, s is an upper bound of the expected number of non-zero elements in $\tilde{K}$.

Following the idea of importance sampling, the optimal sampling probability $p_{i,j}^+$ for $K_{i,j}$ should be proportional to $(P_\epsilon^*)_{i,j} C_{i,j}$ from the perspective of variance-reduction. However, $(P_\epsilon^*)_{i,j}$ is unknown beforehand, and thus $p_{i,j}^+$ is impractical. Fortunately, there exists a natural upper bound for such a sampling probability. Based on the marginal constraints on P_ϵ^* , we have $(P_\epsilon^*)_{i,j} \leq a_i$ and $(P_\epsilon^*)_{i,j} \leq b_j$. Moreover, we focus on the general scenario where the ground cost between supports is bounded, i.e., $C_{i,j} \leq c_0$ for some constant $c_0 > 0$. Therefore, we have the upper bound

$$(P_\epsilon^*)_{i,j} C_{i,j} \leq c_0 \sqrt{a_i b_j}.$$

Such an inequality motivates us to use the sampling probability

$$p_{i,j} = \frac{\sqrt{a_i b_j}}{\sum_{1 \leq i,j \leq n} \sqrt{a_i b_j}}, \quad 1 \leq i, j \leq n. \quad (16)$$

Algorithm 2 Spar-Sink algorithm for OT

- 1: Input: $a, b \in \mathbb{R}_+^n$, $K \in \mathbb{R}_+^{n \times n}$, $0 < s < n^2$, $\epsilon, \delta > 0$;
 - 2: Construct $\tilde{K}$ according to (15) and (16);
 - 3: Compute $\tilde{P}_\epsilon^* = \text{SINKHORN}(\tilde{K}, a, b, \delta)$ by using algorithm 1;
 - 4: Output $\widetilde{\text{OT}}_\epsilon(a, b) = \langle \tilde{P}_\epsilon^*, C \rangle - \epsilon H(\tilde{P}_\epsilon^*)$.
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3.2 Theoretical Results

Without loss of generality, we assume the supports of a and b are identical and ϵ is relatively small. Then, the cost matrix C is symmetric, and the resulting kernel matrix $K = \exp(-C/\epsilon)$ is positive definite.

Theorem 1 Under the regularity conditions (i) $\|\mathbf{K}\|_2 \geq n^\alpha/c_1$ for constants $1/2 < \alpha \leq 1$ and $c_1 > 0$, and the condition number of $\mathbf{K}$ is bounded by $c_2 > 0$, (ii) $p_{ij}^* \geq c_3 s/n^2$ for $c_3 > 0$, and (iii) $s \geq c_4 n^{3-2\alpha} \log^4(2n)$ for $c_4 = 8/(c_3 \log^4(1 + \epsilon))$ and $\epsilon > 0$, as $n \rightarrow \infty$, the following result holds with probability approaching one that

$$\widetilde{\text{OT}}_\epsilon(\mathbf{a}, \mathbf{b}) \leq \text{OT}_\epsilon(\mathbf{a}, \mathbf{b}) + c_5 \epsilon \sqrt{n^{3-2\alpha}/s}, \quad (12)$$

where $c_5 > 0$ is a constant depending on c_1 , c_2 , c_3 , and c_4 only.

Consider a general case that $\|\mathbf{K}\|_2 = O(n)$, i.e., $\alpha = 1$, condition (iii) indicates us to select $s = \tilde{O}(n)$ elements to construct the sparse sketch.

Consider the computational cost of Algorithm 2. Constructing the sketch $\tilde{K}$ requires $O(n^2)$ time, and such a step can be naturally paralleled. The matrix $\tilde{K}$ contains at most s non-zero elements, and thus calculating $\tilde{K}v$ and $\tilde{K}^T u$ takes $O(s)$ time. Therefore, the overall computational cost of Algorithm 2 is at the order of $O(n^2 + Ls)$, which becomes $O(n^2 + Ln)$ when $s = O(n)$.

3.3 Simulations

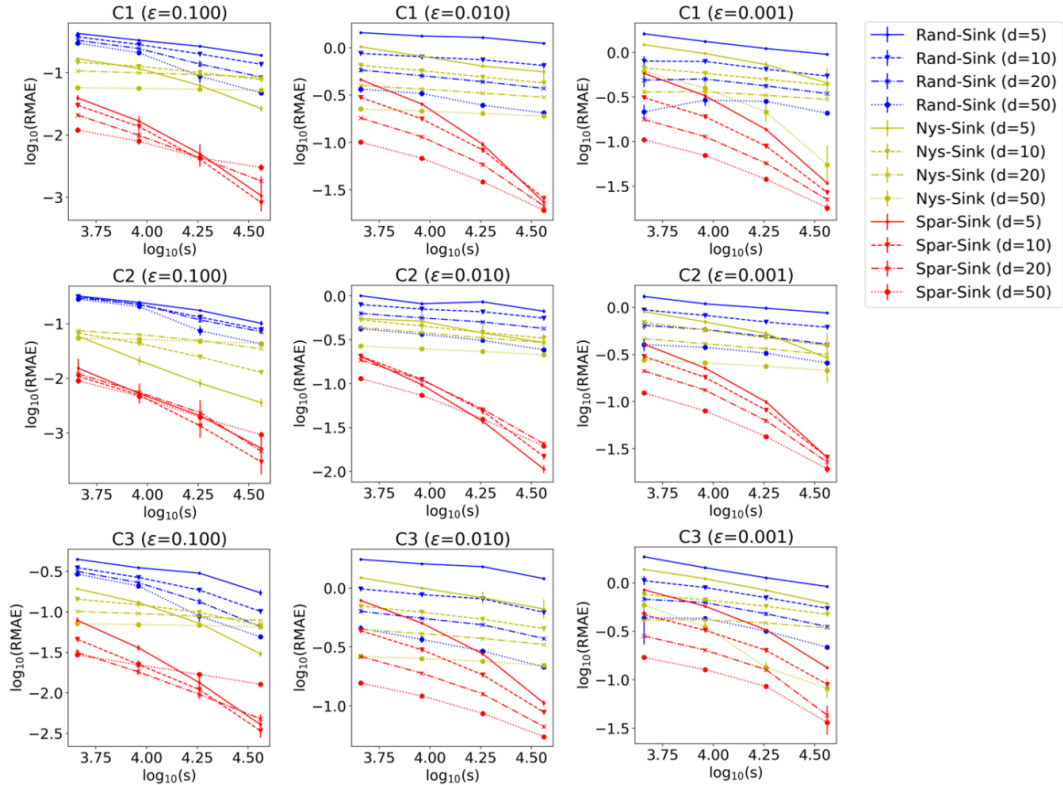


Figure 4: Comparison of subsampling-based methods w.r.t. RMAE(UOT) versus increasing s (in log-log scale)

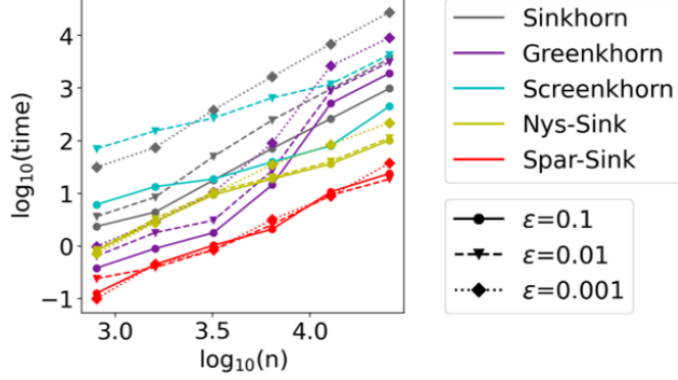


Figure 5: Comparison of different methods w.r.t. computational time (in seconds) for estimating the Wasserstein distance under C1.

3.4 Applications to Color Transfer

The objective is to transfer the color of an ocean sunset image to an ocean daytime image, as depicted in Fig.6(a). The pixels of each image can be represented as point clouds in the three-dimensional RGB space. Due to the large number of pixels in each image, which is nearly a million, we randomly downsample $n = 5000$ pixels from each image, resulting in $\{x_i\}_{i=1}^n$, $\{y_j\}_{j=1}^n$ and use discrete uniform distributions to define $a, b \in \Sigma_n$. We construct the cost matrix C using pairwise squared Euclidean distances, that is, $C_{i,j} = \|x_i - y_j\|^2$. To generate a new image with source content and target color, we compute the optimal transportation plan between a and b and extend the plan to the entire image using the nearest neighbor interpolation.

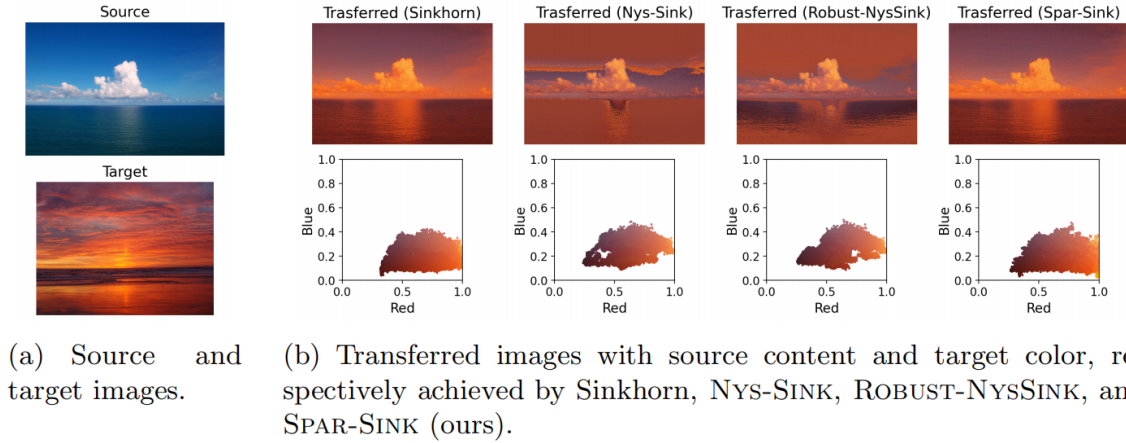


Figure 6: Comparison of different methods on color transfer.

The results of the transferred images generated by these methods are presented in Fig.6(b). Among the evaluated methods, we observe that Spar-Sink produces a transferred image that closely resembles the result of Sinkhorn, and its corresponding RGB scatter diagram is also similar to that of Sinkhorn. In terms of computational efficiency, the CPU time of computing the plan is 60.45s (Sinkhorn), 12.92s (Nys-Sink), 27.74s (RobustNysSink), and 3.15s (Spar-Sink), respectively. These results demonstrate the effectiveness of SparSink on this common computer vision application.

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